Azure Meetup
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RETELIT GROUP

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INTR3

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Workloads are ready for Take-Off: Secure your migration journey with Azure Migrate

Luca Torresi
Zeno Testoni

#GLOBALAZURE 2024

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- Cloud Engineer @ SoftJam Spa
- Mi occupo di: Microsoft Azure, Microsoft 365 e Security
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Agenda

- I prerequisiti della migrazione verso Azure
- Azure Migrate, il servizio e le sue feature
- Sicurezza dei Workload in Azure
- Demo

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I prerequisiti della migrazione verso Azure

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Prima di migrare, fermati e pensa

- Conosco e o ho documentato i processi del mio datacenter?
- Conosco e ho documentato le mie isole applicative?
- Il mio personale IT è stato adeguatamente formato?
- Il Business, quindi tutta l'azienda è d'accordo?
- Siamo disposti ad accettare dei nuovi paradigmi?
- Abbiamo stabilito un Budget per un POC?
- Una volta completata la migrazione, chi governerà Azure?
- Il mio Business può essere spostato completamente in Cloud o devo passare ad un modello Hybrid-Cloud?

Tutti i piani funzionano...



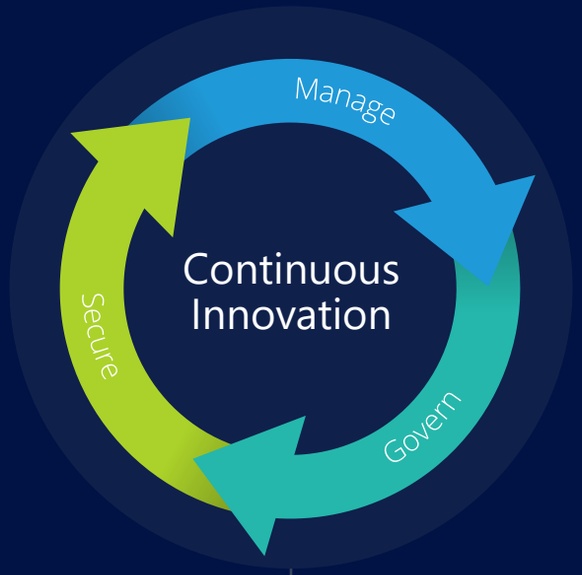
Fino al primo assalto

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BY COMMUNITY • FOR COMMUNITY
2024

Cloud Adoption Framework



Cloud Adoption Framework



Manage

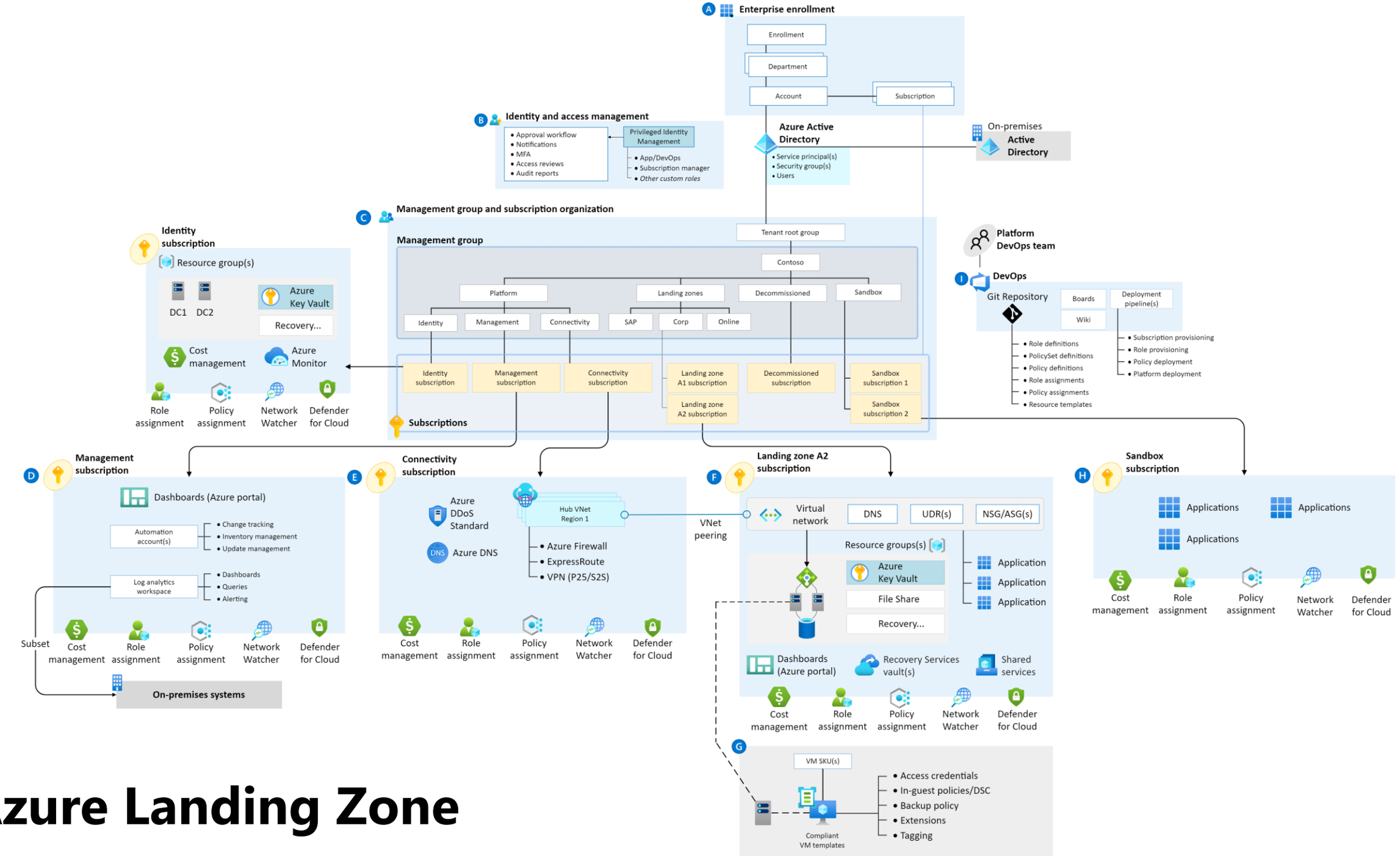
Gestione Opertativa
Definizione delle Baseline

Govern

Gestione dei rischi
Policy e Compliance

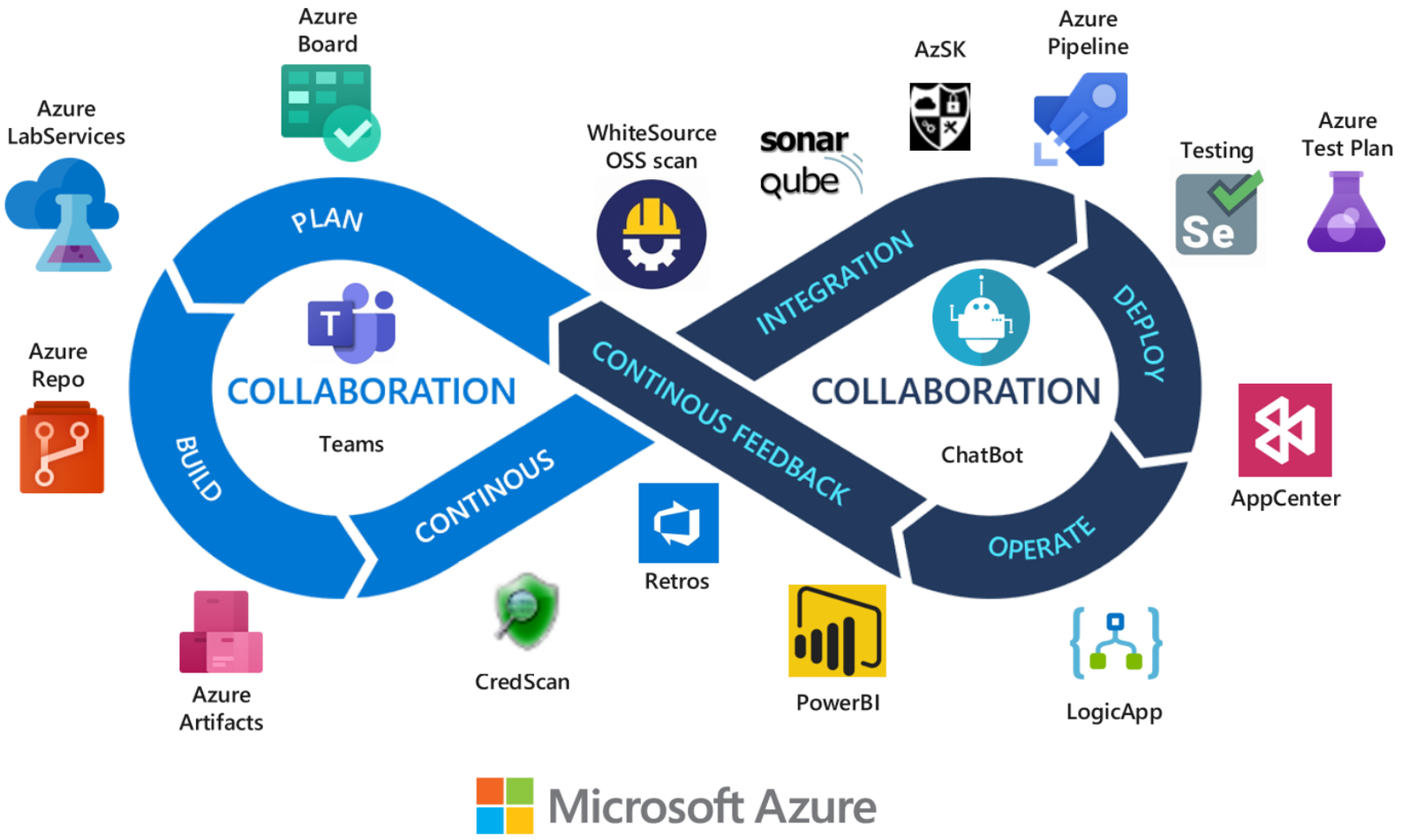
Secure

Messa in sicurezza continua
Protezione di tutte le risorse



Azure Landing Zone

Dev(Sec)Ops

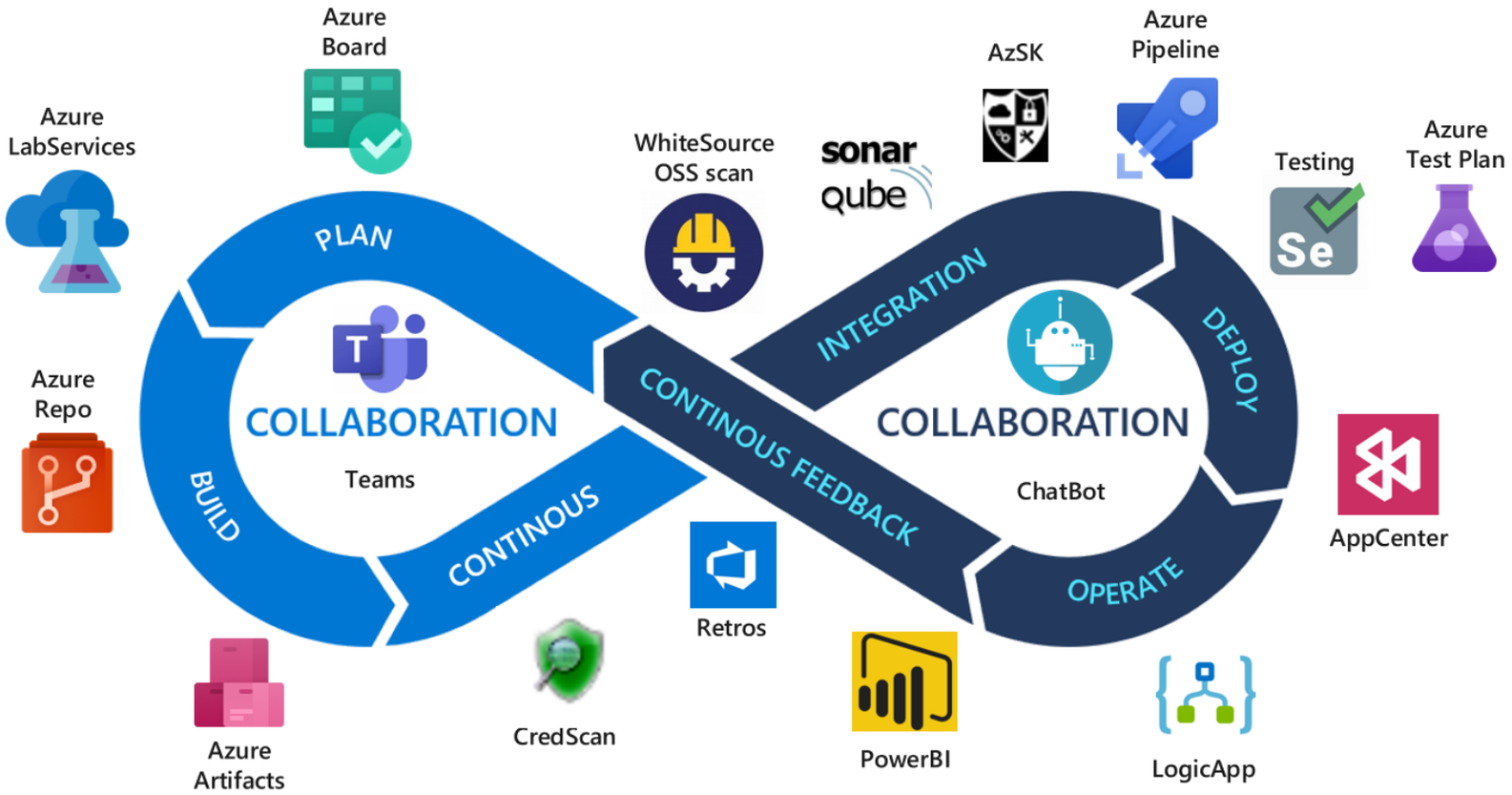


Azure DevOps

GitHub Enterprise

Microsoft Azure

Dev(Sec)Ops



ARM Template

Bicep

Terraform

Microsoft Azure

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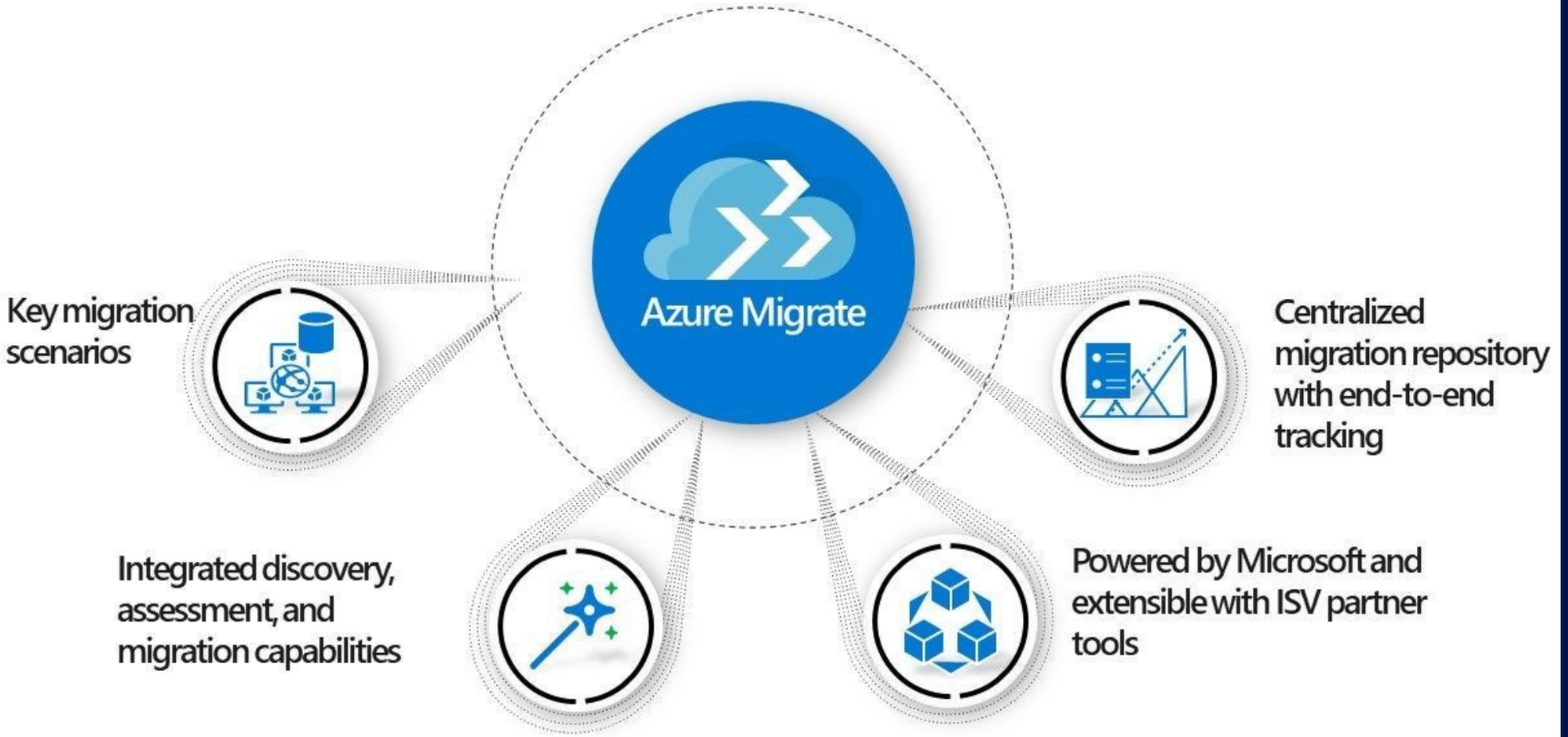
Azure Migrate, il servizio e le sue feature

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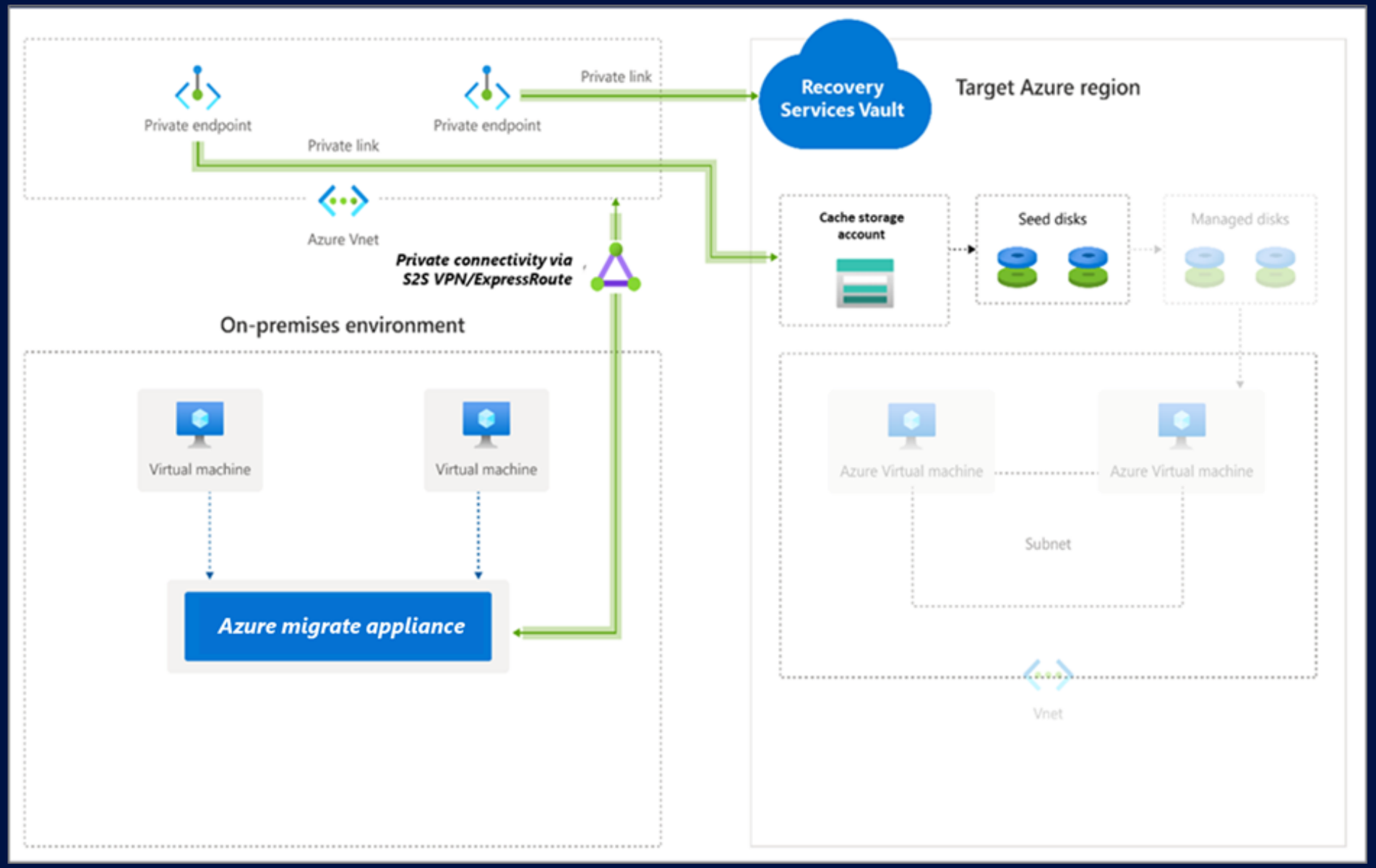
#GlobalAzure

AZURE MIGRATE

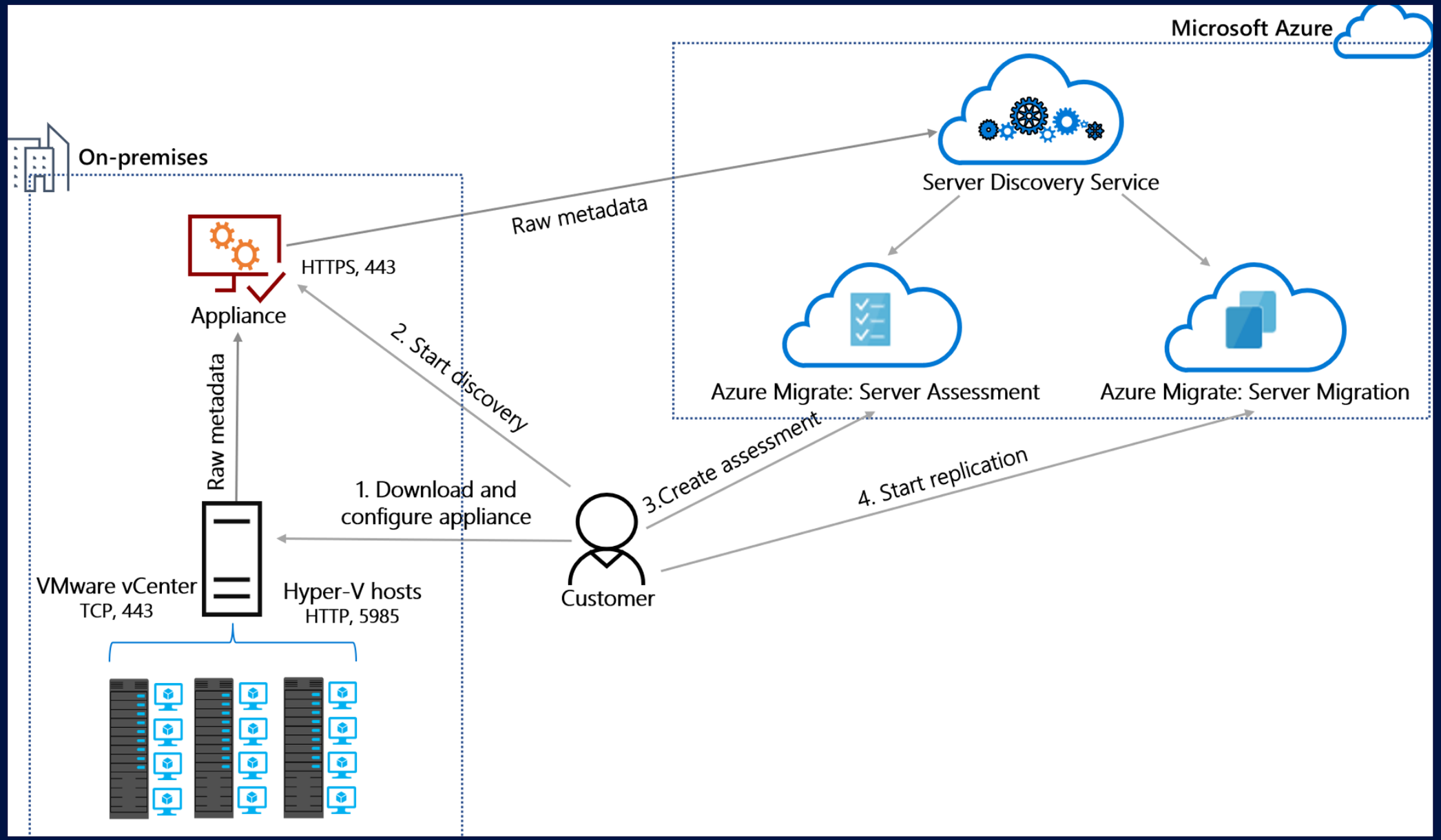
Azure Migrate – Hub for Datacenter Migration



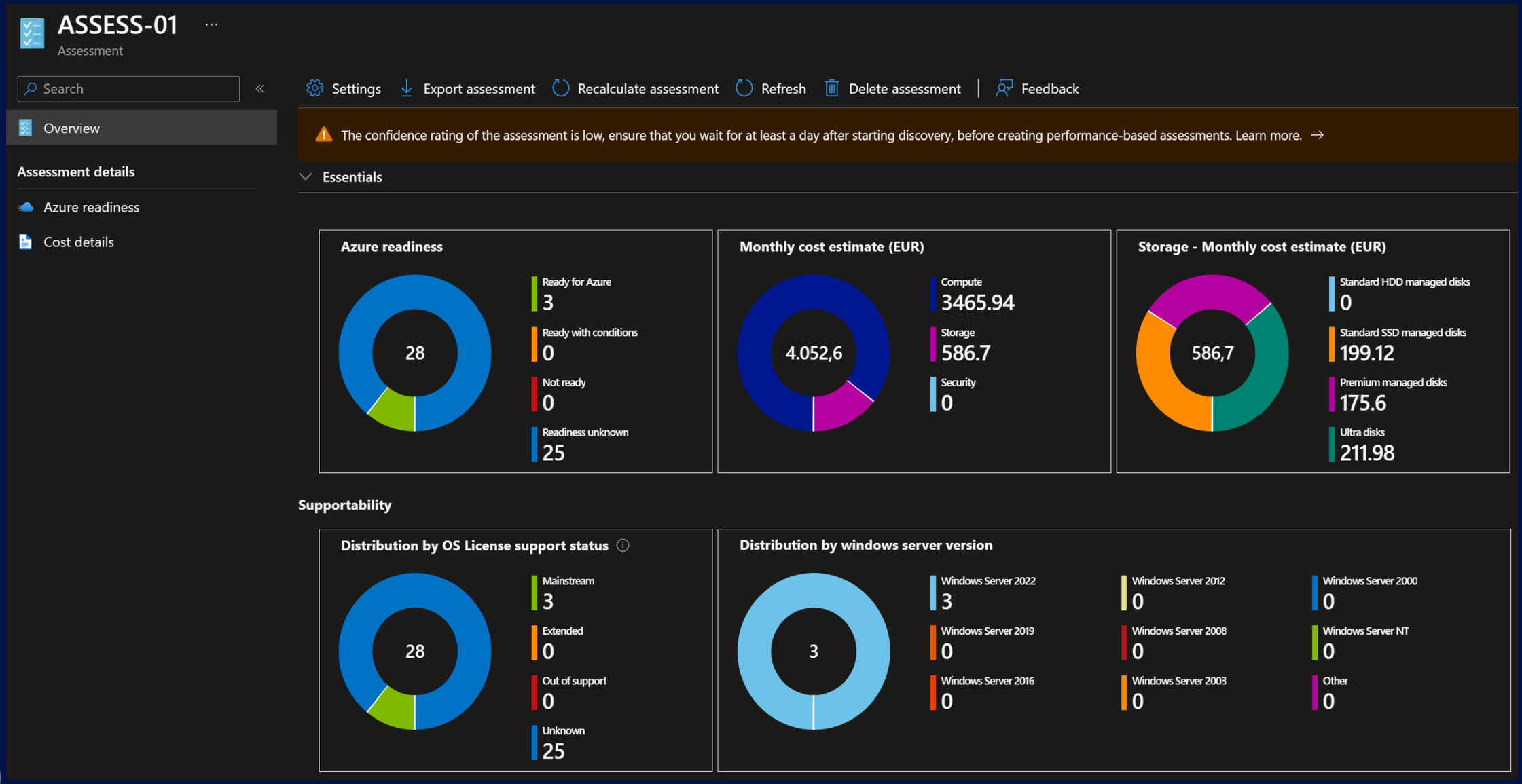
Public or Private Link



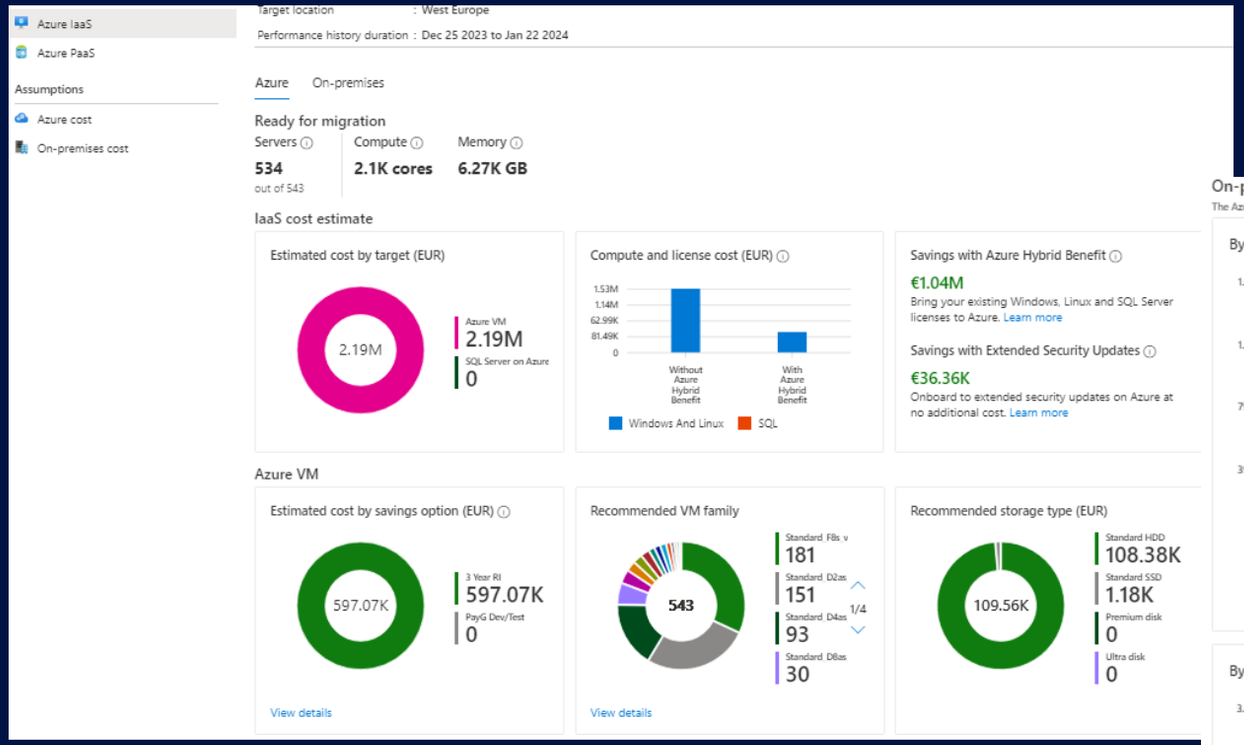
Discovery

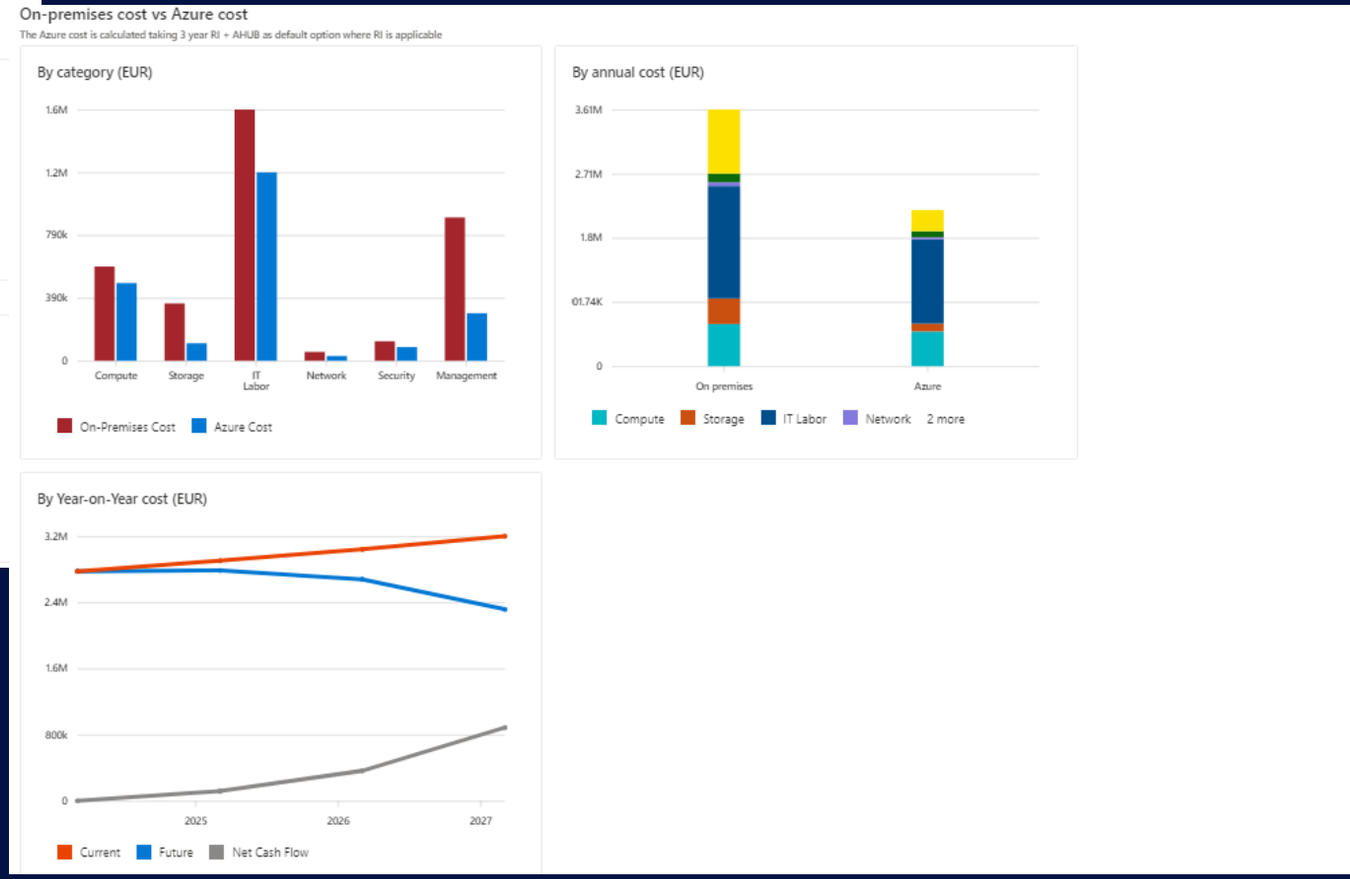

Microsoft Azure
On-premises
Server Discovery Service
Raw metadata
HTTPS, 443
Appliance
2. Start discovery
Raw metadata
Azure Migrate: Server Assessment
Azure Migrate: Server Migration
1. Download and configure appliance
3.Create assessment
4. Start replication
Customer
VMware vCenter
TCP, 443
Hyper-V hosts
HTTP, 5985

Assessment


ASSESS-01
Assessment
Search
Overview
Assessment details
Azure readiness
Cost details
Settings
Export assessment
Recalculate assessment
Refresh
Delete assessment
Feedback
The confidence rating of the assessment is low, ensure that you wait for at least a day after starting discovery, before creating performance-based assessments. Learn more.
Essentials
Azure readiness
28
Ready for Azure
3
Ready with conditions
0
Not ready
0
Readiness unknown
25
Monthly cost estimate (EUR)
4.052,6
Compute
3465.94
Storage
586.7
Security
0
Storage - Monthly cost estimate (EUR)
586,7
Standard HDD managed disks
0
Standard SSD managed disks
199.12
Premium managed disks
175.6
Ultra disks
211.98
Supportability
Distribution by OS License support status
28
Mainstream
3
Extended
0
Out of support
0
Unknown
25
Distribution by windows server version
3
Windows Server 2022
3
Windows Server 2012
0
Windows Server 2000
0
Windows Server 2019
0
Windows Server 2008
0
Windows Server NT
0
Windows Server 2016
0
Windows Server 2003
0
Other
0

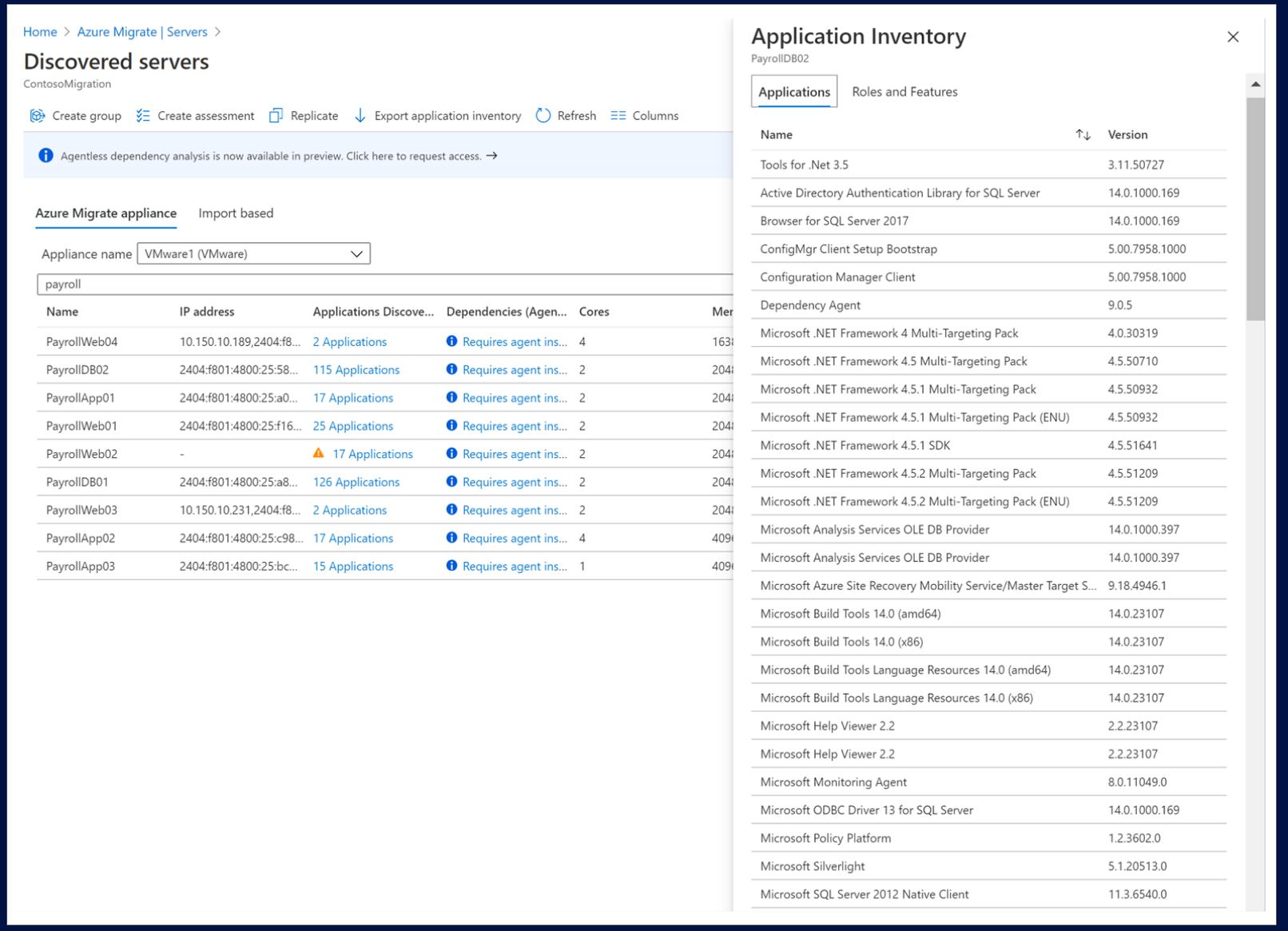
Creazione Business Case





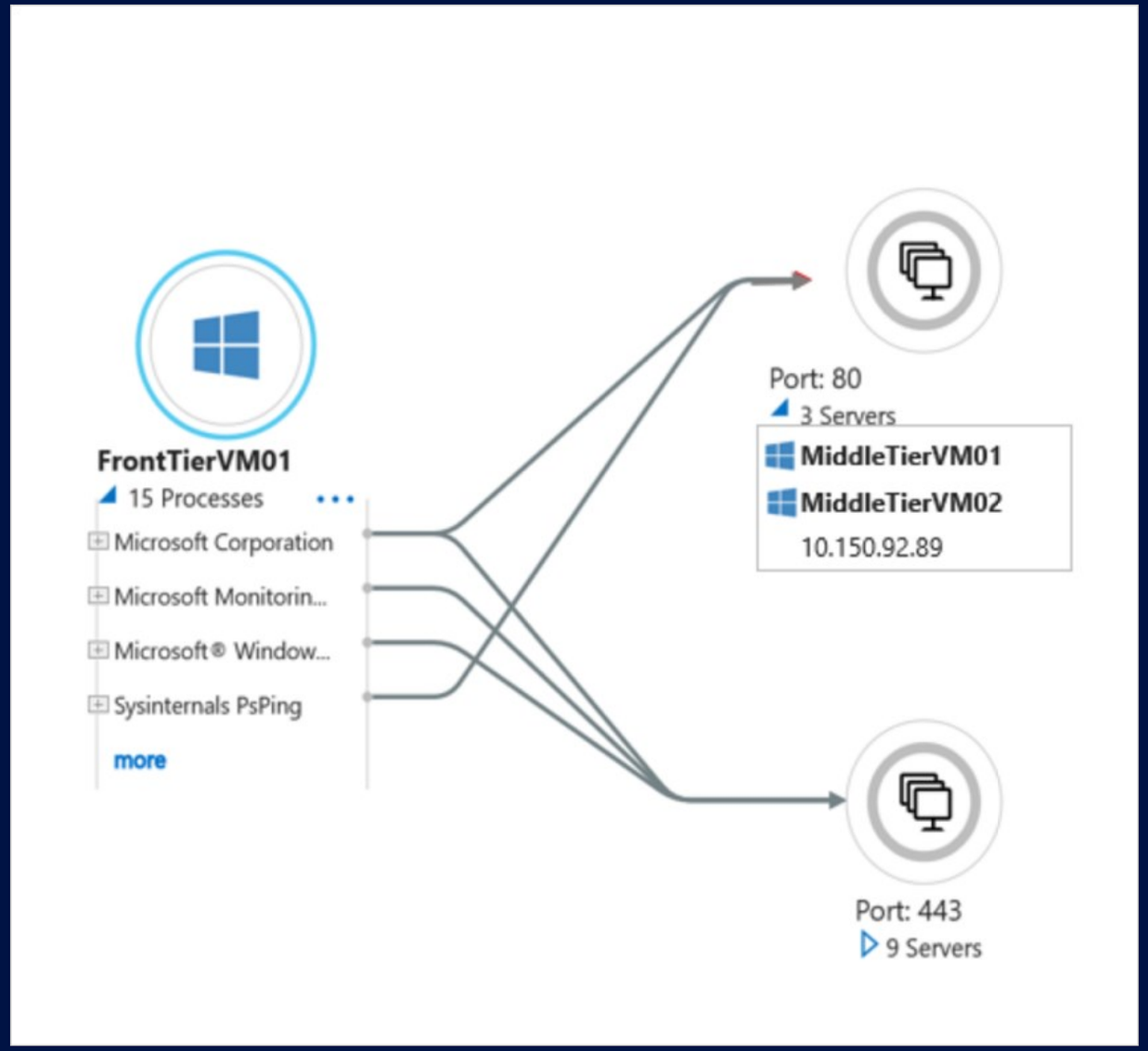
Server Discovery

La server Discovery consente di recuperare tutte le macchine virtuali, le loro applicazioni e i Workload che girano al proprio interno.

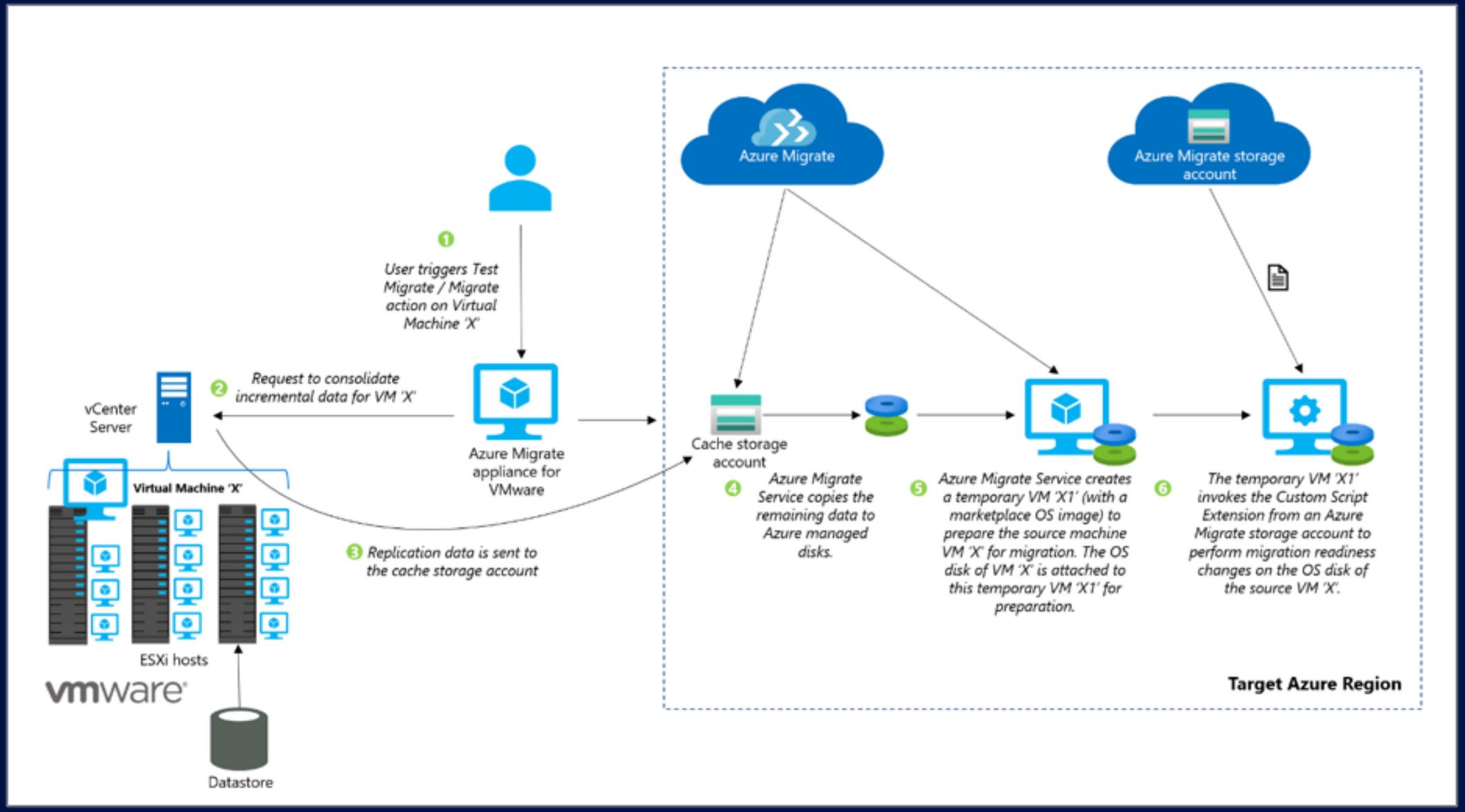


Dependency Analysis

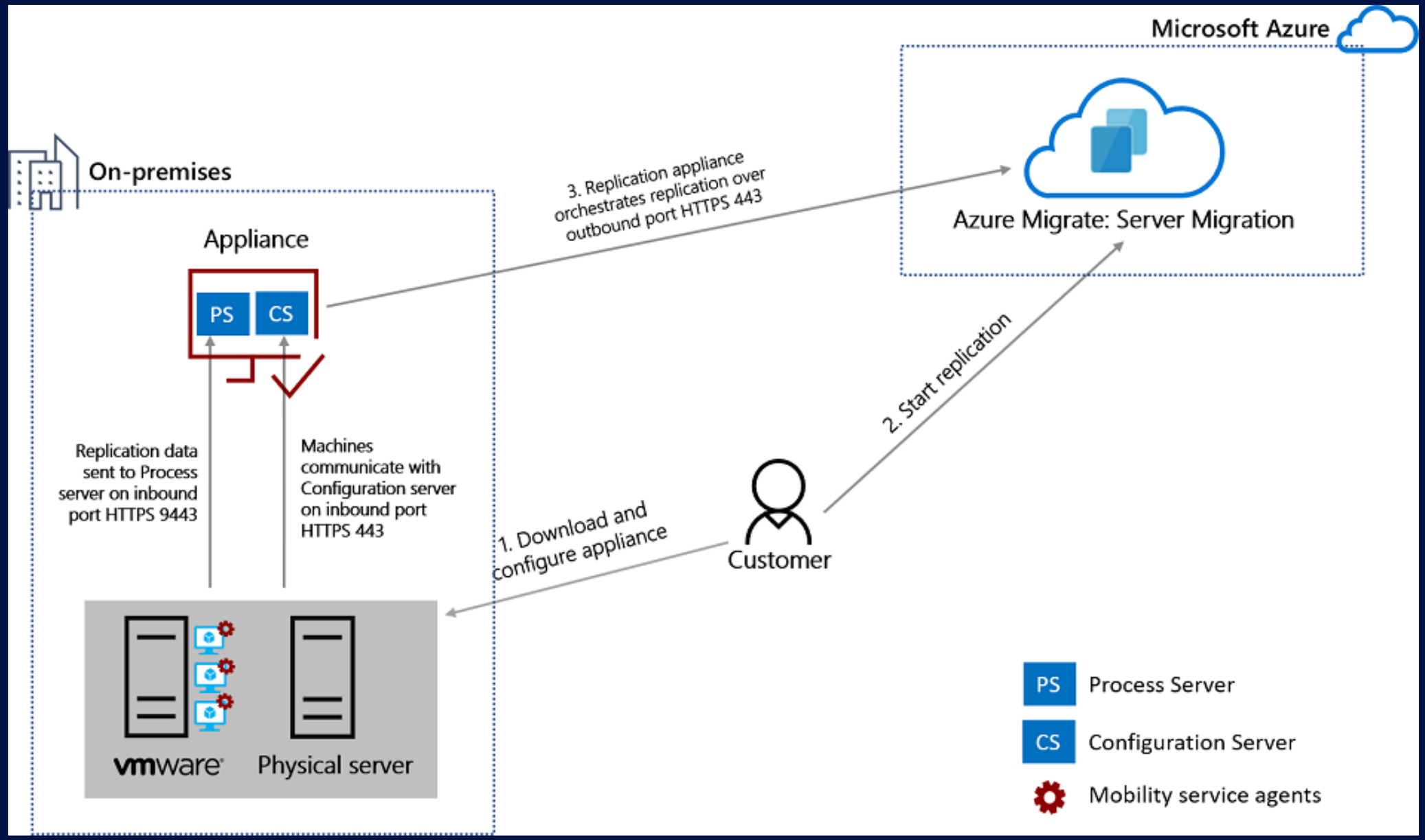
L’analisi delle dipendenze consente di avere a colpo d’occhio una visuale sulle isole applicative e più in generale sulla comunicazione che esse instaurano tra di loro.



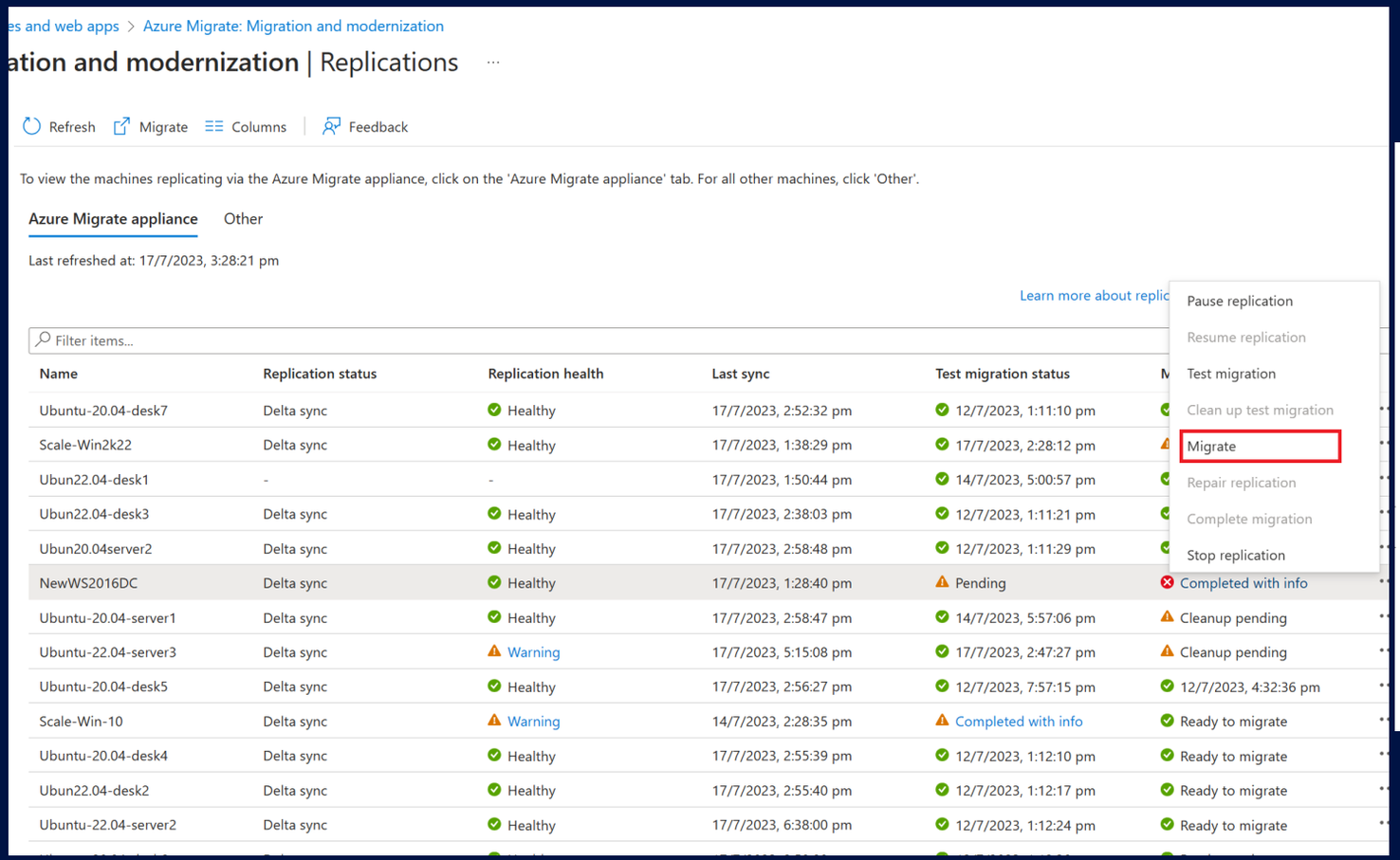
Agentless Migration

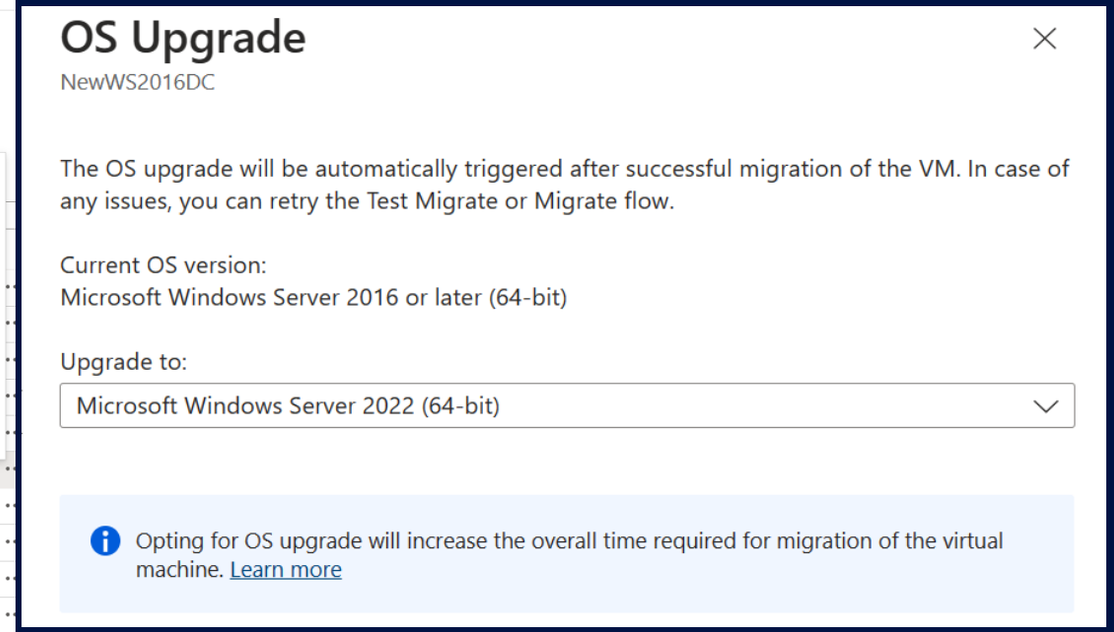


Agent Based Migration



Windows OS Upgrade During Migration





DEMO

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Sicurezza dei Workload in Azure

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Microsoft Cybersecurity Reference Architecture

Security modernization with Zero Trust Principles

December 2023 – aka.ms/MCRA

Security Operations / SOC

Microsoft Security Experts
Defender Experts | Detection and Response Team (DART)

Managed Security Operations
Using Microsoft Security

Microsoft Defender XDR

Unified Threat Detection and Response across IT, OT, and IoT Assets

Incident Response | Automation | Threat Hunting | Threat Intelligence

Microsoft Security Copilot (Preview)

Microsoft Sentinel
Cloud Native SIEM, SOAR, and UEBA

Cloud	Endpoint	Office 365	Identity	SaaS	Data	OT/IoT	Other
Azure, AWS, GCP, On Prem & more	*Workstations, Server/VM, Containers, etc.*	*Email, Teams, and more*	*Cloud & On-Premises*	*Cloud Apps*	*SQL, DLP, & more*	*devices*	*Tools, Logs, & Data*

This is interactive!

1. Present Slide
2. Hover for Description
3. Click for more information

Security Guidance

1. Security Adoption Framework
2. Security Documentation
3. Cloud Security Benchmarks

Software as a Service (SaaS)

Microsoft Defender for Cloud Apps
- *App Discovery & Risk Scoring (Shadow IT)*
- *Threat Detection & Response*
- *Policy Audit & Enforcement*
- *Session monitoring & control*
- *Information Protection & Data Loss Prevention (DLP)*

Microsoft Entra Internet Access

Identity & Access

Conditional Access – Zero Trust Access Control decisions based on explicit validation of user trust and endpoint integrity

Endpoints & Devices

Unified Endpoint Management (UEM)
Intune | Configuration Manager

Microsoft Defender for Endpoint
Unified Endpoint Security
- Endpoint Detection & Response (EDR)
- Web Content Filtering
- Threat & Vuln Management
- Endpoint Data Loss Protection (DLP)

Hybrid Infrastructure – IaaS, PaaS, On-Premises

Defender for Cloud – Cross-Platform Cloud Security Posture Management (CSPM)

Secure Score
Compliance Dashboard

On Premises Datacenter(s) *3rd party IaaS & PaaS* Microsoft Azure

Extranet
- NGFW
- Edge DLP
- IPS/IDS/NDR
- Microsoft Entra Private Access & App Proxy *Beyond User VPN*

Intranet
- Azure Arc
- Azure Stack

Azure Marketplace

Express Route

Private Link

- Azure Firewall & Firewall Manager
- Azure WAF
- DDoS Protection
- Azure Key Vault
- Azure Bastion
- Azure Lighthouse
- Azure Backup
- *Security & Other Services*

Information Protection

Classification Labels

Microsoft Purview
Information protection and governance across data lifecycle

Discover → Classify → Protect → Monitor

File Scanner
(on-premises and cloud)

- Data Governance
- Advanced eDiscovery

Compliance Manager

Microsoft Entra

- Passwordless & MFA
  - Hello for Business
  - Authenticator App
  - FIDO2 Keys
- Entra ID Protection
  - Leaked cred protection
  - Behavioral Analytics
- ID Governance
- Microsoft Entra PIM
- External Identities

Defender for Identity

Active Directory

Securing Privileged Access – aka.ms/SPA

Entra Permission Management – Discover and Mitigate Cloud Infrastructure Permission Creep

Privileged Access Workstations (PAWs) - Secure workstations for administrators, developers, and other sensitive users

Security Posture Management – Monitor and mitigate technical security risks using Secure Score, Compliance Score, CSPM: Defender for Cloud, Microsoft Defender External Attack Surface Management (EASM) and Vulnerability Management

Windows 11 & 10 Security

Network protection	*App control*
Credential protection	*Exploit protection*
Full Disk Encryption	*Behavior monitoring*
Attack surface reduction	*Next-generation protection*

IoT and Operational Technology (OT)

Azure Sphere

Microsoft Defender for IoT (and OT)
- *ICS, SCADA, OT*
- *Internet of Things (IoT)*
- *Industrial IoT (IIoT)*
- *Asset & Vulnerability management*
- *Threat Detection & Response*

Defender for Cloud – Cross-Platform, Multi-Cloud XDR
Detection and response capabilities for infrastructure and development across IaaS, PaaS, and on-premises

Defender for APIs *(preview)*

People Security

Attack Simulator | Insider Risk Management | Communication Compliance

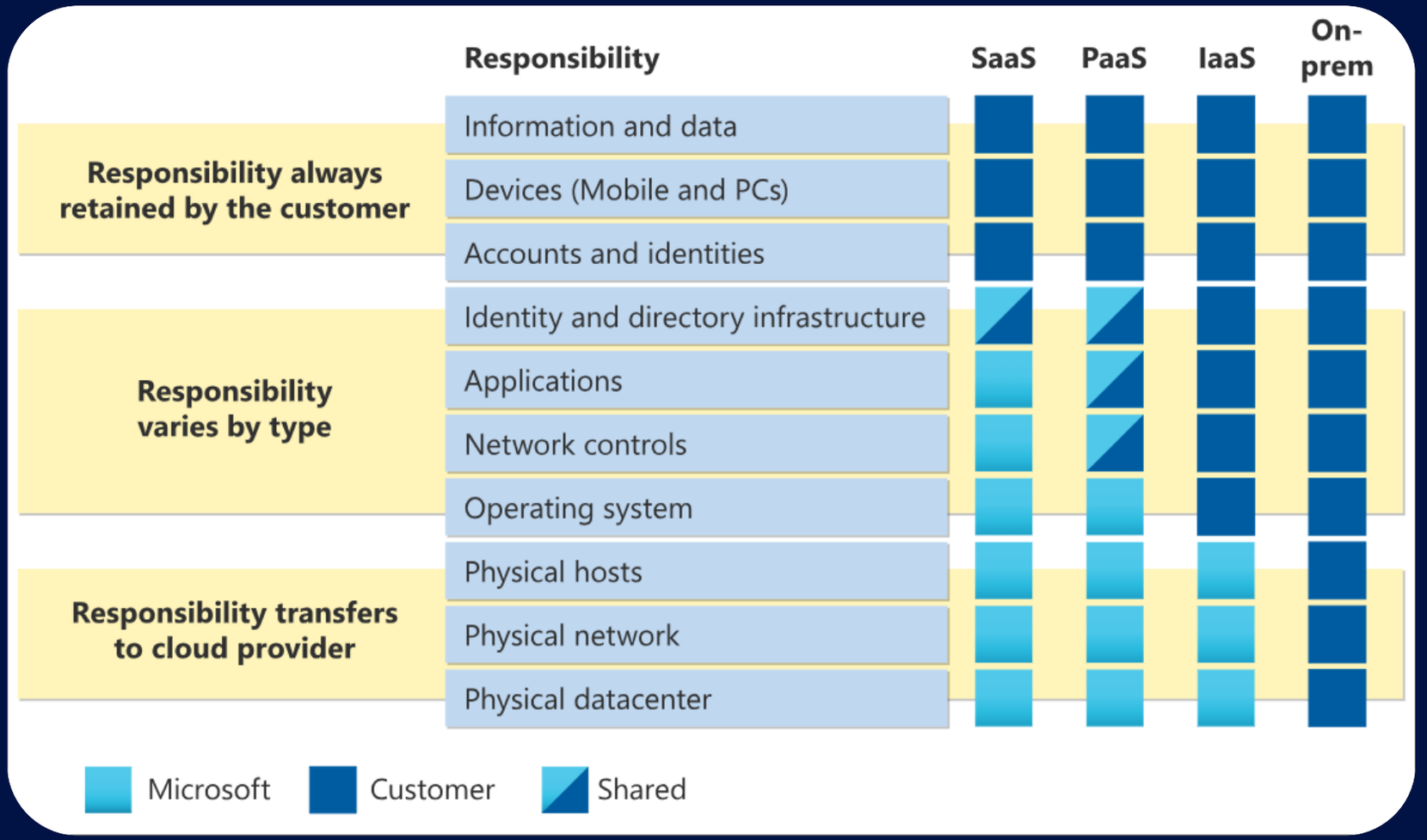
GitHub Advanced Security & Azure DevOps Security
Secure development and software supply chain

Threat Intelligence – 65+ Trillion signals per day of security context

Service Trust Portal – How Microsoft secures cloud services

Security Development Lifecycle (SDL)

Everyone takes responsibility … please!



Zen…. Ehm …Zero Trust !

Verify explicitly Use least privilege access Assume breach

Some tips!

- ✓ Workloads are monitored and alerted to abnormal behavior.
- ✓ Every workload is assigned an app identity—and configured and deployed consistently.
- ✓ Human access to resources requires Just-In-Time.

- ✓ Unauthorized deployments are blocked, and alert is triggered.
- ✓ Granular visibility and access control are available across workloads.
- ✓ User and resource access segmented for each workload.



Let's get serious !

Define your business needs:

- Does your company need to assess and strengthen the security configuration of its cloud resources?
- Does your company want to manage the security posture of multicloud resources from a single point (single pane of glass)?
- What boundaries do you want to put in place to ensure that your entire organization is covered, and no areas are missed?
- Does your company need to comply with industry and regulatory standards? If so, which standards?
- What are your goals for protecting critical workloads, including containers and servers, against malicious attacks?
- Do you need a solution only in a specific cloud environment, or a cross-cloud solution?
- How will the company respond to alerts and recommendations, and remediate non-compliant resources?
- Will workload owners be expected to remediate issues?

Let's get serious !

.... And define an adoption strategy

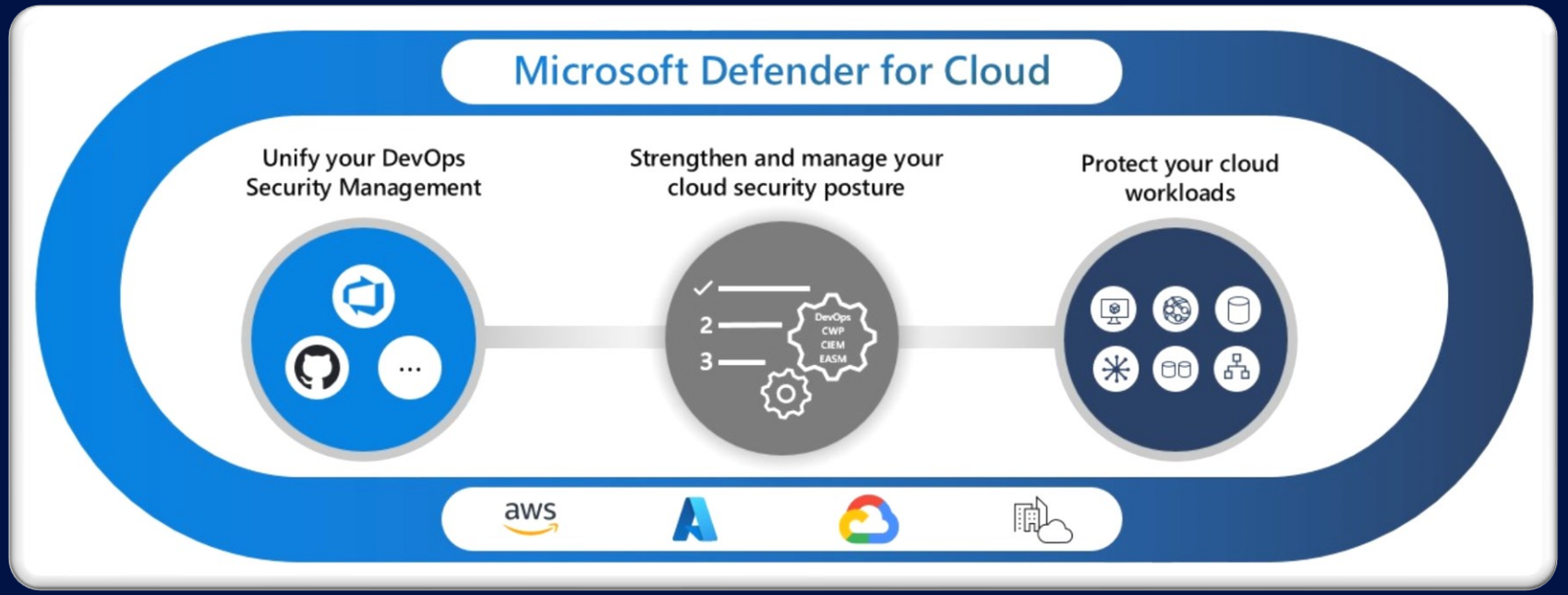
Centralized model - Security controls are defined, deployed, and monitored by a central team.

The central security team decides which security policies will be implemented in the organization and who has permissions to control the set policy.
Workload owners on the other hand are responsible for managing their cloud workloads but need to follow the security policies that the central team has deployed.

Decentralized model - Security controls are defined, deployed, and monitored by workload owners.

Security control deployment is done by workload owners, as they own the policy set and can therefore decide which security policies are applicable to their resources.
The security team usually has insights into the overall security posture of the organization, and they might hold the workload owners accountable for improving their security posture.

Defender for Cloud



Defender for Cloud

Cloud Security Posture Management (CSPM)

Microsoft Defender CSPM provides advanced security posture capabilities including agentless vulnerability scanning, data-aware security posture, the cloud security graph, and advanced threat hunting. Pricing is based on subscription size, with billing applying only for Servers, Databases, and Storage resources at $5/Billable resource/month.
Foundational CSPM includes asset discovery, continuous assessment and security recommendations for posture hardening and a Secure score which measure the current status of your organization's posture.

Plan	Pricing*	Resource quantity	Monitoring coverage	Status
Foundational CSPM	Free Details >		Full	Off On
Defender CSPM	$5/Billable resource/Month Details >	0 resources		Off On

Cloud Workload Protection (CWP)

Microsoft Defender for Cloud provides comprehensive, cloud-native protections from development to runtime in multi-cloud environments.

Plan	Pricing*	Resource quantity	Monitoring coverage	Status
Servers	**Plan 1** ($5/Server/Month) Change plan >	0 servers	Full Settings >	Off On
App Service	$15/Instance/Month Details >	0 instances		Off On
Databases	Selected: 0/4 Select types >	Protected: 0/0 instances		Off On
Storage	$10/Storage account/month $0.15/GB scanned for On-Upload Malware Scanning (config Details >	0 storage accounts		Off On
Containers	$7/VM core/Month Details >	0 container registries; 0 kubernetes cores		Off On
Key Vault	$0.25/Vault/Month Details >	0 key vaults		Off On
Resource Manager	$5/Subscription/Month Details >			Off On
APIs	**No plan selected** Change plan >	0 Azure API Management services		Off On

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What we learned ?

Se conosci il nemico e te stesso, la tua vittoria è sicura.

Se conosci te stesso ma non il nemico, le tue probabilità di vincere e perdere sono uguali.

Se non conosci il nemico e nemmeno te stesso, soccomberai in ogni battaglia.

Sun Tzu – L'arte della guerra

DOMANDE



GRAZIE

